Multenions and Differential Invariants.

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§ 1. Preliminary Explanations. -Since multenions appear almost been designed by Nature to serve as an algebra dealing with such matters as differential invariants and relativity, I have thought it desirable at the present

moment to present a summary of their properties. Towards the end propose to enter into more detail in applying them to differential invariants, and to point out some of the special properties of multenions when n, the vector complexity of the system, has the particular value 4.

Clifford first, I believe, applied the subject to elliptic space of dimensions. His bi-quaternion q + ayr, where co 2 = 1, is the general member of a sub-algebra of multenions when n = 4. In the form of multenions suitable to deal with real questions of Belativity Hamilton's bi-quaternion q-\-rx/(- 1) presents itself in a precisely similar manner, and of this I shall have more to say towards the end of the paper. Joly's ' Manual of Quaternions has its last chapter devoted wholly to multenions. I believe tl elaborate attempt at development is my own paper in *

* Proc. Boy, Soc, E 1907-8, p. 503.

The notation used below is based as closely as possible on quaternion notation. It is practically identical with Joly's so far as the latter goes.

differs in several respects from the notation of my own former paper. changes are all in the direction of simplicity, and back towards quaternion rules. The same applies to changes in terminology. These changes, which I believe are all for the better, are in the main due to friendly advice received from Prof. Knott in connection with the former paper.

Law I. Besides the unit scalar 1 there are given n generators, which will generally be regarded as representing wunit mutually orthogonal vectors in a

Multenions and Differential Invariants.

Euclidian space of n dimensions, i\, i2 , ... i n .
ordinary laws of algebra except the commutative law of multiplication apply
to the general integral expression q, called a multenion, where

where every x is a scalar. Since any x may be negative we understand this
statement to include the ordinary algebraic uses of the signs + and
we do not understand that any assertion has been made about the use of the
sign -r- [q-i-r can later be understood to be an alternative for qr" 1].

Law II. Scalars are commutative not only wi
each of the % primitive unit vectors. [Thus scalars are c
multenions.]

Law III. [i x 2 means iiii, etc.]
In- i
f\
i>\
a
- i
ft
02
&
~- i

• . .

Law IV. In a product, called a primitive vectorium, ia%..., of primitive
unit vectors, if two adjacent different vectors be interchanged, the product
merely changes sign.

Such are the assumptions. The rest is development, aided of course by
suitable terminology and symbolisation.

Def. 1. A primitive unit means any one of the following
(2) any primitive unit vector ia , (3) any other primitive vectorium
[Strictly (3) includes (2) and in a sense (1).]

Def. 2. An expression of the form p - 2 x i where each x is a scalar, is
n
c =l
called a vector, or a Vi or a n Vi,

Def. 3. Any multenion is obviously capable of
2,xi a ib . . . i/c where ia %,... 4 are m different primitive vectors and m ranges
,

from to n, m being given Ymq means the sum of all parts of the multenion for that value of m. Thus

q = V 2 + Vi2 + V 2+...+Vn 2

Y a q is said to be the part of q of homogeneity a and is called a Y a or a nYa or a homogeneous multenion or a hyper vector.
to invite suggestions for a settlement of the name.
"an a- vector" but this clashes with relativity
6-vector. In multenions with n = 4, there are two kinds of 4-vector
and 4V3.]
Y q will generally in words be called the scalar part of q but in the present connection we must not use Bq as an alternative for Y In applications we

very decidedly need to be able, in one system of multenions (n = 4) to make

no change in the quaternion notation but yet
notation and the full multenion notation as applying to one symbol q.
hq = V # + V 4 $

Endless confusion would result from allowing V to be represented by S.
Def . 4. If uh #2 , ... ot a are any ^-vect
vectorium. [Hence we have already called such an expression as i\i2H a
primitive vectorium.]
Def. 5. A linear multenion function of a multenion is called a multenion
Unity. Ya Unity, etc. An extended vector
Similarly for a vector linity or a n
linity is a special kind of multenion Unity, <l>, defined from a given vect
linity, <fi.
Its fundamental property is this : If a 1} ot 2> ... oc a are any ^-vectors

whatever, a itself ranging from to n

[It is not obvious that this remark covers a possibility in general.
definition, which will be modified below,
importance of vectorium s and extended Unities in our subject.]

Usual Notatio

Scalars (V) : a, o, c, g, h, Jc, I,
9 (= arc), ir.

General multenions :
p, q, r y s.

Hyper vectors (homog. mult.) and vectoriums

V<as

Homogeneities a

Hyper vectors u

c
Vectoriums a

(These conventions should be noted and remembered.
of formulae by their property of minimising number of suffixes, etc.)

Vectors (Vi) : a, /S. 7, 8, e, £ gf, rj,

Linear funcbions ; |, 77, A,, /^, z/, m v,

Vectoriums : a, e, ^, t, 0, p, £, co.

Differential symbols : S, A, H, L, d, *#/?, ^5,

Selective Unities, etc. : K, P, Q, E, S, T, U, V.

Def. 6. Complexes of vectors, hypervectors, multenions, etc., are sufficiently illustrated by multenions. If q h q 2j ...,

multenions of the form a X x qG are said to form a co c

G-l

can be made linearly dependent on h and no fewer, then h is called the complexity of the complex. Thus the complexity of the complex of all vectors is n, of all multenions it is 2 n , of all multenions of he

Def. 7. A unit vector or a unit vectorium is one, a, such that a = +1.

[The square of every vectorium is a scalar, but u2 is a scalar in general only in the four cases when a = 0, 1, w - 1 or w.] Thus all the pr are unit vectoriums.

Def. 8. Two vectors, a, j3, are said to be normal to one another when Voct/3 = 0. Two vector complexes are said to be completely normal to one another when every vector of the one complex is no the other.

Every one of the symbols Ya is a multenion linity. They are all commutative with one another. They are all idem-potent, that is Ya2 = Ya and they

all satisfy the equation VaV& = when a =fch.

There is another system of multenion Unities of great utility and simplicity. They are 2 n in number. They are all commutative with each other and with each of the n+1 Unities Ya . Let Pa be that definite linity of multenion, q, which simply reverses the sign of every ia which occurs in the expression (1) for q. From the well-known properties of Unities a product of the Fs such as P1P2P3 simply reverses the sign of every ii, i2 and is. Unities are (1) unipotent, that is, if

<f>

= PJP& . . . Pa,

then <fi 2 = 1 ; (2) they are proscriptive, by which is meant that

<f>

(qr) = (f>q

<j>r,

where q and r are any two multenions. It is convenient to use product P1P2 . . . P?i of all the Unities P a so that , P reve primitive vector.

K is defined, consistently with its quaternion use, in the following manner. ~Kq is a linity of a general multenion, q, which reverses the sequence of the primitive unit vectors in every primitive vectorium I, of q, and also reverses the sign of every primitive unit vector. Thus

K(%^3) - (-iz)(-i2)(-~ii) = iiisfk-

K is (1) unipotent, that is K = 1 2

; (2) it is commutative with every every P; (3) it is retroscriptive, that is K(gr) = X.rKq prove it when q and r are two primitive vectoriums and then generalise. Such a method of proof is very often applicable and may be referred to as

proof by reduction to primitive units.]
But from these properties not only is

K (qr) = E> . Kq,

but if <jf> be given by (4), K<£ (= <£K)] is also retroseriptive. if <£, yjr, % ..., are any number of multenion linities, every one either proscriptive or retroseriptive, then also is the product,

it is proscriptive when the number of proscriptive or retroseriptive ;

^, %, .,

which are retroseriptive is even, and it is retroseriptive when that number is odd.

The special retroseriptive PK (= KP) is of sufficient importance to have a symbol appropriated to it. We shall call it Q. Thus the product of the three P, Q, K is equal to the third.

A caution to users of present methods in relativity and hyperbolic geometry seems desirable. It relates to the discussion of real questions imagery

J = v/ (~l).

Let us illustrate by anticipating our mode of treatment of real questions in relativity. We begin with the multenion system based on the primitive generators i lf i2 , is, i4 . Then we put

u = J*»

and drop out of most of our subsequent work all reference to the original i 4 and J 2 using i instead with its property

, i

2

= 1, as opposed -to %

Questions in relativity are then treated by aid of the multiplicative combinations of such expressions as

xi\ 4- y&2 + 2*3 cU y

whereby, so long as all the scalars thereby introduced are real, dealing with real questions in relativity. The general multenion constructed contains 16 and not 32 independent real scalars.

relativity purposes real. We shall call such a multenion system semi-real.

Now return to the meaning above assigned to K.

and also K (iii 2 ist) = HHHh

but whereas K(iii2 nu) = (^iW^)~\

it is not true that the corresponding equation holds when we replace u by i. I have therefore refrained from making the definition of K depend on such equations as (9). It is, nevertheless, a perfectly easy theorem to prove that

Kl = t\

when /is any product of our original generators i\, i2f ... in This ceases to

be true when i takes the place of i±, simply because J" 1 is -J and not + J.

It may be noted, however, that there is always a reproscriptive K<£ be found, which is such that for every semi-real vectorium l Hi = l" 1 3

Thus, suppose we replace i a+1> ... i n by 4 +]

where i = J 4 and therefore
c = 1, c = a-\-1, a+2, lc
2

Then K = Kl «+i 1 «+2 *» = Qiril 2 I • • .

is easily shown to have the required characteristic.

We now obviously have

Yu = (-)"?/,, Qu = (-)i«(«- 1)^,

Further if we put
Sc = V<#, c = 0, 1,

Tgr = ^ - ^ + ^3 - 23+...

= i0- + p)2 = £o + £2 + 24-K.. >,
P(o)2
p (d2 = K 1 - p)2 = 21+23+25+ •••_,
Q2 = + yi- 22- 23+24+26- ..., ffo

K2 = 2o"-2i - 22 + 23 + 24-~25-...::

Clearly, we might introduce Q Qi, and K Ki, corresponding to P(, Q,
All the linities recently considered (Vc ,
capital linities.

§ 2. Theorem of Independence and

q q2 = -222b then #1 and #2 are said to be anti-commutative.
x
following enunciation, a product of a differen
2b 2& • • • %m is said to be odd-formed or even-formed, according as a is odd
or even.

If 2b 2s> ••• 9.™ oe m multenions such that qi

differing from zero, and each pair is anti-commutative ;
2 m multiplicative combinations are linearly independent ; and, if
are independent unless the product q\q 2 ..qm of all of them is a scalar, and, if it .

m ~ l and only 2 m_1
is a scalar, 2 of the combinations are independent, and these
~l
2m independent combinations may be taken as the odd-formed combinations, or instead as the even-formed combinations, or instead as the combinations from which one assigned mulienion, such as qm ,
linities as well as of multenions.

Thus, if n is even, a general multenion cannot be expressed by fewer than
2 n
independent scalars. In fche case of n
fewer than sixteen independent scalars.
expressed by four independent scalars, the same
n = 2. [But to express it thus, we have to impose the condition that i\i2H
is a scalar such as - 1. This is practically
Quaternions.] Either of these cases might be r

Quaternions. I prefer to view Quaternions from another side
forming the subalgebra of even homogeneities for the case n = 3.

In general multenion theory, it is convenient deliberately to prescribe for the case n odd that the 2 n primitive vectoriums shall be independent. This is equivalent to saying that the product v = i\%2 ... primitive vectors is defined as independent of the scalar n is odd.

Whatever be the value of % multenions with even homogeneities only ~1 form a subalgebra with 2 n units. When n is odd, this subalgebra can always be identified with the next lower multenion system.

When n is even, the number of independent scalars, 2 n is a perfect square. ,

In this case the algebra is the equivalent of an algebra of Unities with the same number. The characteristic equation, therefore, satisfied by a multenion, may be found. When n is even, the degree of the characteristic equation is 2*» When n is odd, the degree is 2^ n+1 \

In the case of n = 4 or 3, the characteristic function may be written in a variety of allied forms, of which the two most important are

chfq (x)=E(q-x)’K(q-x) . (2Y - 1) [(#- x)K(q- x)]

= (q-x)Q(q~x) . (2Yo-1)[(q-x)Q(q-x)1

The coefficient of every power of x is a scalar which it is h fn write down in full. Suppose these scalars are m, m’, m \ m

chfq {x) = x* - m’ " x^ m"% - m’x + m.
-\-
2

This defines the scalars m. We now have the identity
q - m "q + m"q - m’q-\-’rriE:0>
f
4: 3 2

In the case of the semi-real system suitable to relativity, c

positive or zero, never negative. The positive value of its fourth root is the natural generalisation of the quaternion tensor -\-\/(q^q)- uncertain whether to denote this positive scalar by Tq, thus

Tq=+{qKq.{2Y -l).(qK S)}i\
= +{- 2 Q 2 .(2Vo-l).(ffQ?)}*J

An objection to doing so is that this is not Hamilton's meaning in the case
+ Jr. His tensor is then imaginary, namely, one of the
of his biquaternion q

values of [(q + Jr)K (q 4- J>)]i In our form below, this amounts to saying
that the tensor consists of a V part + a Y part. 4

It suffices for our purpose to consider that q only has a tensor when qKq is

a scalar (V). For our semi-real relativity case this must be a real

and in that case Hamilton's meaning, and the meaning given by (5), coalesce.
[If qKq is a scalar, then

2 K 2 = (2V,,-1), 2 K 2

and therefore Tq above = {(^K^) }*.] 2

The general value of q for which qKq is a scalar is
q = xeP where (1 + K)jp = 0.

The following theorems will now be easily proved by any reader slightly familiar with quaternions.

q = tiYoi-iq « S^Vo*?,
7?, 71

/o = 2 iV 4 V = S i ^ov,
c =
=1 C

V (2 V» = V (rVtf) = V VtfV.r),

Vogr = Y rq,

whence the cyclic form

Vo (qRr) = Y (rB 2),
V (jjKBr) = V (rKEj).

Here E is any capital retroscriptive and KE any capital proscriptive.
Thus, for the capital Unities, not only are they all commutative, but they are all self-conjugate. Conjugacy is denned and tested
Formally, if <f>
is a given multenion linity, its conjugate,
linity for which
V #r = Vorpq,

q and r being arbitrary. When = <£ <£',
generally, for any <£, !($ + <£') is the self -conjugate part
is the skew part. More general kinds of conjugate present the
occasionally. For instance, let E be any one of the capital retroscriptives.
QR ,
may be defined by Yo^E^r = V rR<£ R
the E~eonjugate of $, With

these meanings of the conjugate, <£', and the B-conjugate, <p R of we have

(^ + ^)r = ^R + ^R, (^)n = ^R^R,

^)ji = B<£'B, (/>' = B<£rE

The general meaning of a conjugate, <f> c>

of

a unipotent retroscriptive linity of the linity

function of the linity $: = $ (2) retroscrip

(1) unipotent because ((/>')'

because (<^)' = ^'<jk' ; (3) linear because (<£+-^)' = + I have be me in MS. a full analysis of the properties of such a conjugate, but must not give it here. [E^ stands to q as <j>' stands to self -B-conjugate as well as self-conjugate. If (f>q <f>'q

= rqp, <f> nq = R,p

and similarly if <j>q = piqn +P2qr2 +

Frequently, as in proving the above, we require

if Yopr =* V gr, then p = q,

r being arbitrary, and again

if Y ap = V /3p, then a = (17)

/o being arbitrary, and similarly for a n Ya instead of a general multenion or a vector.

When in a semi-real system E is chosen so that "Rl = Ir

have that if q = Ixl and #' = 2a?'/, then

Y qRq = 1xx\ f

and in particular Y$q&q = Ix 2

§ 3. The Commutation Theorem.- If. la is of homogeneity a, and 4 of homogeneity 6, and if there are exactly c primitive vectors common to the a vectors of la and the b vectors of 4 it is easy to see that a + b- 2c, and that 44 = - ( )

ab ~~ c

hL- Hence

Va+b-teUV = - y*~ x Ya+b-teVU. (1)

(

[Obviously true when u = 4, v = 4 as we see by considering separately the two cases x = x^ It is therefore true for u and

c, c. This is a case of

proof by reduction to primitive units.]

(1) may be called a first form of the commutation theorem as it shows us a

first effect of commuting u and v in uv. (1) at once
called the full form

UV = (Y a+b + Ya+b-2 + + V - OT 1
. . . ffl 6)

= (-)«*(Va+6 -V -2+... + [-] Va -)m L a+ft

= (-)^ + *(V a - -V a . 6+3 +... + [-] Va+6)wJ
ft

The following are easy deductions

| [flW + (- a6w] = (Ya+b + Va+6 - + Va + 6-8 +
) 4

|- [W - -) ^] = (V,, + &,,2 + V a + 6-6 +
(
tt&
UV

and these may be written in the reverse order with (- yb+b in place of { -) ab

From (3)
l(U(D + (OU) = (Ya +2 + Ya -2)Ma) = (V« + + Va ~)ft>^
| (%ft)
- ft)?6) = V UCO = - Va COW
.
h .

which has important applications to infinitesimal rotations.
cases are

u2 = (+ V 4 + V 8 +))uv/2 . .
. ,

© a = (Vo + V 4)© 2 ,

/3 = (V + V 2)/3 = (V - V 3 0*,)

V «/3 = -V /8a,
2 a Vo^/8 = Vo/3#.

V */3 = £(*£-#*),
2 Vo«/3 =
| (a/3

§4. Botations of Three Kinds. - Suppose we find n multe
such that for all purposes of Laws I to IV they may take the place of i\, i2 ... in

are we to call the substitution of the multenions p
rotation ? In Euclidean space of three dimensions we should not always so affirm. If i, j, k should represent, as sometimes in quaternions they do, three directions in a rigid body we could not move the body so that i became - i and j and k remained unaltered; but they would continue
conditions of the laws. We should call the transformation a perversion or reflection rather than a rotation. As the algebra of multenions is (n even) a full linity algebra, or (n odd) a sub-algebra of such a Unity algebra, we will assume that given q,
q˜ l can be obtained uniquely in general so that
qq˜ = 1 = q˜\ (though in certain singular cases we have to say instead that
l

go can be found so that qq = = qoq). In the general case we will say that q is vertible (i.e., it has an inverse q"); in the singular case that q is non-
1

vertible.

Theorem concerning multenion rotation. (1) When q is a g
multenion, then in the four fundamental laws the primitive vectors i\, i 2

may be replaced respectively by

= 2h2~\ Pn = 2M""

1

Pi •• > ,

and the effect of the replacement is to change any multenion r to

r

f

= qrq" 1

.

(2) Conversely if p\, p 2 ...p n are n multenions such that

- 1 ±p *zsp£=

1 ... =p n2 ,

and every pair is anti-commutative, that is

P1P2+P2P1 = 0, etc.,

and when n is odd

PlP2-..Pn = ilH*~in,

then there exists a vertible multenion, q,for vjhich (1) and (2) are true ; r
meaning the midtenion obtained from the given multenion r by replacing
k> i2,.~in ty pi,p2, ...p n respectively.

This kind of rotation is scarcely what we seek.
in general leave V unchanged, that
c is, it is

qYcr.q-i^Ycizrq- 1

Multenion rotation qrq" 1
is operation by a multenion linity

such that $’</> -1= <f>(j>’,
and conversely.

Next, imposing the condition that (6) is to be satisfied, we find that q must be the product of any number of vectors, and that if
is satisfied. We may next find what is the effect of infinitesimally changing the vectors and attempt to build up the finite continuous group of which that infinitesimal transformation is the element.
transformation effected by changing q to q + ^coq - qo)), where co is infinitesimal and the corresponding finite rotation is e^Oe" where oo is finite

w is a perfectly arbitrary nY2 infinitesimal of course for the first case. When
,

eo is not singular (co
2
= 0) its general form may be put
CO = cl€\€2 ~\- ^€364

where ei, e 2 , e3 , e4 . . . are orthogonal unit vectors

e™ ’ss (cos a + €\€2 sin a) (cos 6 + €364 sin b) ...
and the factors here are commutative. Puttin

e°> = ei (ei" cos a + € sin a) e (e^ cos h + e sin 6)
1

2 3

we see that e w is the product of an ^m number of vectors. The general form of co when it is singular is not simple, but I have a complete solution before

me in MS.

To show that the infinitesimal rotation is as stated will not occupy too much space. Suppose

r' == a/3.. A r X" ...^"

. .

1

where the vectors a, yS, 7, ... are assumed to be unit vectors, as that does not affect the transformation. Let a, /3, ... change to ct + da, /3-M/3, ..., still remaining unit vectors.

dr' = d(prp^ - dpp" 1

)

1

That c^p"" 1 is a aV2 is straightforwardly proved thus

^p-i _. daoC ljra . ^/3/S""

1

. a^-f a/3

Every term here is a W V2 ; for example, consider the third term.

y + dy are unit vectors, Yodyy" = 0, and therefore afyy"' is a nV2 It follows

1 1

that the third term is a W V2 If we put u, v! for r, r' and J doo for dp^r

1

.

efo' = -J-

(dmf/ - u'cldd) = Yod

This is the very simple, very useful form in which true infinitesimal rotation

presents itself in our present subject. A rotation of vectors p i
(ftp
of p such that <£'<£ = 1.

When without qualification we speak of rotation we shall understand it to mean e ( )e˜ 0> bi
(a continuous group).

The skew part of any vector linity, <£, is of the form of an infinitesimal rotation. Thus, introduce the f notation of quaternions,* by which

n
*(?,$)= 2 F(i,i),
c =l

where F is bilinear in its two members. We have
p-=-rv &

f P =-?v ^v= -rvop^g:

The skew part is
J(*-f)P = iV!(V 2 ^)p = Vi©/i.

Here we have assumed that
Vi (V**£) 7 = a V0i87-/3Vo7a,

much as in quaternions. This is nob difficult to prove, and we shall in § 7 enunciate much more general theorems of which this is a particular case. The pure part does not similarly reduce. We have
i(^+*)p=-i(rvo/»^+^?v p?).
/

§ 5. Vectoriums. Let

Then in the expression Va + 6+c (a^>/3 7 (6) (c)
it is obvious
primitive units that within the brackets we may at will write either
Va a (o)
, and /3 (6) or Vb/3®, and 7 (c or Vc7 (c>; the value of the expression is
>

unaltered by such insertion or removal of Ya Yb, or V The statement, of , c

course, is in general only true of this particular part, Ya +i+ of the product

of highest homogeneity. Next, it is similarly evident that if we change, say, a 2 to xu 2 + yoL2 we simply alter the corresponding vectorium similarly, that is

Again, if in the vectorium we interchange two consecutive members such

as u2 *3, we merely change the sign of the whole. For we ma
write instead of #2<*3

2 *3,
Ot, Y 2 Ci2^3, -Y2*3*2, -a a 3

Finally, by a familiar simple process we may interchange any two members by successive interchanges of neighbours, and as a final result, again, the vectorium changes in sign merely. Thus Ya ^a) is "com

* See ' Utility of Quaternions.'

common simple properties of a combinatorial mul
probably well known to my readers and may be assumed.
rently first, in a perfectly general manner, appreciated the value of, and fully
established these properties. Indeed, Y a o^ a)
is both algebr
metrically identical in meaning with Grassmann's combinatorial " product."

should like for a second time to express
. my unbounded adm
Grassmann's great pioneer work, though I cannot help believing that the
actual form of the matters he dealt with has been put out of court by later
developments.]

We may note the following theorem in passing. I
Grassmann's products. If p is a vectorium of homogeneity, g, then

Va + b + c(Yg+a+b pUV)(Vg + c Wf>) = Ya +b + c (Yg + apn) (Yg + b + c VWp).

Of u, v, w the middle one, v, may be passed over from the first to the second
bracket-pair.

The following is wanted immediately and is constantly in request.

be thought of as a passage into or out of the operation of a Y

pre-faetor or post-factor] of either v or p n We have .

vYa q = Yn-aty, Y a q v = Y n - a (gv). .

Also, if p n is any multenion of homogeneity, n p n = xv, and, therefore, in the 9

present process, p n behaves exactly as does v, that is

PnYaq = Yn - ap n q, Yaq p n = Yn - a
.

Suppose there are n given linearly independent vectors
«(a> be the product of any a of them, and d n " a the product
)

such sequence that Vn aMw ˜ a always retains the same value.
<x(>

h = Y n -a0t^- a) . Yn^a^a^-^)'
[By a convenient, strictly speaking indefensible,

˜
really = oo , we mean by V n "˜V>W' : c/)

There are 2 n linearly independent [proof of lin
n v
omitted] values of d, and they will be called the 2
#1, *2, ... a n- [The term may, of course, be used when ai, « 2
linearly independent.] The 2 n corresponding values of &,
linearly independent, will be called the normal reciprocals of the vectoriums, a.

They are so called because
Yoda = 1, Yodia = 0,

ai being a second (not corresponding to a) a. [Thus, when

sixteen equations of the type Y dd = 1 and 240 equations of the type
Yo&ia = 0.] Both equations are proved quite simply from (1) by passing
pn = Vw ~V >a<w out of the operation of V in Y aa.
a ~~
a>

Let now q = %xd = %ya. From the normal reciprocal relations we at once
get x = V(#a, y == Yoqd. Hence
a = t&Y q& = %Y a a^ Y n qu^~ a
Q Yn^a^a^^ 1 .
) .

K 2

the last result in the first line coming from a new application of the
theorem. If in (2) we put q = u, getting a more particular result, we shoul
at once see that (2) is analogous to the quaternion theorems

pSotfty = otSflyp + . . . = YftySap +

A particular case (a = 1) of (1) is

= - "Vn -iaia3. ,.a n Yn .
1
aiU2...cc
This is clearly the solution of the problem. G
Sa c a c = 1, c = 1, 2, ...

S«&a c = 0, b^c ; 6, c =

what are the vectors ac in terms of the vectors a c ?
have a precisely similar solution for a c in terms of a c
the relations between the quantities d and the quantities & are symmetrical.
No doubt we have here established this if only it be evident what is the
precise significance of the word " symmetrical " in

this connection. The need of the caution may be illustrated thus. we put d = Ya uiot2 ... <*n

we are tempted to say that a must be Ya did2 ... da but we shall find that we

do not thereby always obtain Bad = 1. The true value of cl is

a = V a aa a a -i...ai = y V a aia2...<xa

when d = Va«i«2...«a

If the ac are taken as the ic , it is obvious that the d vectoriums, and, from the normal reciprocal relations, it the a are also the primitive vectoriums (with different signs in accordance with)

d = Ki

Eeturn to the £ pairs of (7) of § 4. Let multenions. From the definition of £, it is easily seen that

I (V«SiS2...ro,V gifa ../e)=:a! 2Fa,4) = (˜)^ +1) «! SF(4,4-i)

,

fl >

Abbreviate thus

66-6 = £ (a)

,

(-)i"<« +1 Ya

[Thus 0' = 1, 1'= -1, 2'= -1/2!, 3' = +1/3! etc., the signs running
»j«. i

- -|- -j-
"~t~ -r*
"-~ ~~~ ...
.J

Thus SF(4,4-) = «'F(Va ^)
1

From (6) « = SkVowSf 1
= aX£

In (6) put the second = ^V ££W. Thus, V £<°>

a'FW>, V.fM) = a'S(V {<»)Voar< «) = 2F(4, i).

We might have modified the first Va £ instead of the second. (<t

collecting results
a'F W>, = 2F(4, /a"
V,,£<«>)

= 2F(a, a) = 2F(a, a)
(»C a terms in each summation.)

Just as in quaternions F (f, <££) = F (<£'£ f) so now if ^ is a w V a Unity

a'F (V £<«>, £V £(«>) = a'F (</>'V^>, V«?<»>).
fl fl

If $ is a multenion Unity
n n
t a'-F(Va ^"\ <j>Ya ^>) = % a'¥(cj>'Y a ^\ V,,£«>
a= «=

For a - we must put a' = 1, and for each V f (a) we must also put 1 [see (7) above].

§ 6. Extended Vector Unities, -The theorems applications in the multenion treatment

. of

treatment may be considered to begin in § 8, and is explicitly introduced in § 9.

Let <f) be a given vector linity. The extended li which may for most purposes, but not for all, also be denoted by <£, is here defined thus. Let ia - i g ik ... ii be any primitive uni

a itself being arbitrary. Then

<j>YJa - Ya <j>i(,<lyi>h..

ISFow change the n primitive vectors i\, i2 such steps as replacing + yi 2 (or, if i h i2 have already changed

ii by xix

to ex, eg, the change should be from ei to xe + ye2) Such a change simply

multiplies each side of the defining equation (when i\ occurs in meaning is therefore unaltered. It follows that <j>Y

is arbitrary and the vectors a are arbitrary. extended Unities of <j>, ty, we see that ($)(^) <£i/r, for

(<£) (ty) Y aigih Ai = (<p) Yatyigfyih.

. . . .yfri

From this again it follows that the extended linity of ^>" 1 is (<£)˜ 1

Maltenions and Differential Invariants.

Let us use either (<£) or <£ for the extended Unity of cf> for the present. What we have just proved may be put

(4>) (f) = (W)>

We are now about to prove a similar proposition concerning <£ namely

By (7) of § 5 we have

W= (<£')•

or (cj>) u = a'Ya (££)<«> V ^H

Let w, u be two arbitrary multenions of homogeneity a. Operat Y Uo{)

V Mo (<£) « = a'VoWo (<£?)<«> V <H

Modifying the left by the fundament rule of the meaning of (<£)', and modify the right by the rule F (£,<££) = F (<£'£, 5) applicable to a We obtain

Since w is arbitrary, by (16) of § 3

which by comparison with (1) proves the assertion that (0)' = (<£').

The following theorems are also true

= Ya +b<l>u<l>v,

cj>Ya + b ttv

<j>Y a - UV = Ya-bfals&^V,

b

when <£ is non-vertible we must change ^ to <£%. An alter convenient than (3) generally, is thus

$Va-bUfiv - Ya-b^UV,

but this alternative is true without qualification.

v having its usual meaning (i h i2i . . . in) it is easy to see that </> m is the determinant of <£. Putting u - v in (3) and (4) we obtain

v~~

1

6(vv) = md>'~ l v]

or

V = mv

1

v <p(v<p

/

v)

r*

[-4dfcferf February, 1921. -Closely connected with the extension vector operation to multenion operation, of a given vector linity, is a second kind of such extension. If <£ E is the extended (or, as I now prefer t extensive) form of $ of § 6, Its fundamental property, showing its dependence (in a multiple manner) on <£, is that

Similarly, if ty e is the new extensive form of d>, its fundamental property showing its dependence (in an additive manner) on <£ is that

The dependence is called multiple or additive because

(<£3>)e = (jb<l>E, (f + ^)e = f e+^.

We may speak briefly of <£ E as the extensive of <£ and \/r e as the subex tensive of a|r (using extensive and subextensive both compare offensive and defensive; compare also Whitehead's translation of " Ausdehnungslehre " as " Calculus of extension ").

For both kinds of extensive it is true, when </> E a vector, that

fap = fa ^eP = fp,

so that quite frequently it is convenient to use occasions to use -^ for i]r e . [We should very rarely with a given single occasion to use both <£ E and <£ e but if we do we must have some mark to distinguish them, since they have diverse meanings.'

Although the properties of a subextensive linity may be derived directly from its fundamental property defined above, they may instead be derived from those of an extensive by the methods connecting an infinitesimal with a finite transformation. Thus we may put = 1 + $dt = 1 + tydt <£

= B ™ (l-\-<fidt) = l + dt k

(1-f yjrdt) Fj (jf>

Vj

where dt is an arbitrary infinitesimal scalar. Thus from form
once have

Yf = fY - VfV

c e e c c e c,

(*.)' - W)«

tyeVa+bUV = Va+6 (^ V + Ufav), .

ty Y a ~ uv = Y a

e b - fy u v-u\fr 'v), b e .

^r U = l/r/ii = - u VoS^ft

c .

^ = -v ^rVft-i?v - -V^CW? o-fVa+nHH

6 .

i~i^ (kO - -ty/ + V ^K)«> = -V ^fV6+ i^,

e

the last being obtained by putting u = u in (9) above.

Multenions and Differential Invariants.

If $ and i/r are given vector Unities then <j>-E^e4>B~ l is a subextensive that

is it is the subextensive of 4>i^<j>~ 1 - ^ or by (11)
(<j6^<]b
1
)e
/y = - Vft0^ ^b-lty = ~</)E^f^-i^E'

by applications of both the equations [see (2) and (3) of § 6],

If <£ is of assigned dimensions, <£ e is of the same, but not <£ B

far as its dimensions depend on <£ E and not on -^, are a times the dimensions of <£.]

§ 7. Formulw Analogous to the]

Quaternion Formulae

Yafty. - Let a1? ot 2 ..., a a+6 be a+fr given vectors and let a
>

of any a of them and aS the product of the rest in such a sequence that
b)

Ya+iaWa^ always has the same value. Then u being as usual

Note that a< a) occupies the middle position throughout.

o = %- a it easily leads to the last form of (2) of § 5.
u and v being as usual and p any vector,
Vi+ (a~b) uY b+1 pv = Vi+ (0^6) [V a -iup
. + (-)VVa ,, w],
. tf

"Vl+(a.5)(Ya+l^p • ??) = Vl+(a.6)[w"V6_ipV + (-) a /oV . w],

Ya+b-iupv = Va+6 -i[Va -i^ . v + ^Y 6+ i o'y + (

1

-

To make practical applications of multenions it is absolutely essential that facility should be gained in using these formulae. The following comments are to aid in the acquirement of the requisite facility.
work without a list of at least the four equations (4) and (5) before me. Each of those four generates two others. (2) and (3) are those two derived from the first equation (4). If the second term on the right of this e
it (4a)) is removed to the left it can be made to merge with the

equation (2) is produced. Similarly, if the first term on
removed to the left equation (3) is produced. Now as after a li
with (4a) before the eye, the whole of this process can be gone through mentally and the desired modification of an expression under treatment can forthwith be written down, (2) and (3) are strictly speaking not required in the working list. Each of the four equations (4) and (5) is similar in the points described to (4a). The reader who desires to acquire this facility could not do better than begin by writing out all the twelve equations implied.

Attend now only to (4) and (5). If the two equations (4) a
together one reaches a mere identity almost immediately. Therefo
of the equations (4) readily follows from the other. Exactly similarl
of the equations (5) follows from the other. To obtain one of (4) from one of
(5), place v as a post factor to one of (5) and then pass it into th
operators Yx( ) till it reaches v, noting how those operators must be changed
in the process. When v has come to juxtaposition with v, change vv into w
so that n-b=ze and get rid of b 9 leaving a and c. The completion o
verification is simple. I have verified that it is possible to prove eithe

(5) by reduction to primitive units. I have found no simple proof.
I established one of the twelve equations by taking p to be one of the vectors
a c> say a a+b in (1). Perhaps this is the simplest method.

Because the space I have assigned myself does not permit otherwise, I must now omit several chapters in the MS. treatise from which I summarising, but I find one matter requires insertion before I pass to the part of the subject lending itself to applications to differential invariants.

If (f> is a multenion linity its characteristic function chf$ (x) n-ic in the arbitrary scalar symbol x. Another name for it is that it is the determinant of x- 0,and from the point of view suggested it may be denoted by \x- 0|. In our present notation this is given by the identity

chf^x) = |#-0|
= n'Yn &*KYn [(x-<l>)S\M [,
n~
EE x - m\X + ...+(- n y
n l
)
m<)
-n

[V is the same function of n as a' is of a in (5) of § 5, that is n' - (-)% n n+1)Jnl

mn here and in quaternions is generally written as m]. Several forms d
from (6) are useful in the present subject. They are expressions for ma of which it is to be remembered that mn is a specially important case.
(1) a fundamental form in f, (2) a form involving n given linearly independent vectors a h u 2 > ... #n> (3) a form dependent on the general expression for <f>

<£e = - /8iV eai- $2Voe«2-~... = - 2/3V e*,

and (4) the particular case of the last for which

06 = - Voey.

In the order mentioned, these forms are

ma = a Yo(Ya ^cf>Ya ^)
/

= 2 V a [Vo (0a)< V,,- a «<»-«>]
.
a>
. . Yn-iu&al*-'*
r

= % YoYautoKYapW
.

a'Vo(V«V (o) V«€<«>)

The last three are all deducible from the first. Another form could at once be written down from the first, involving the vectors a and the vectors a.

For our applications below the fourth form is important.

F (fc #) = F (vi, ei).

From this V g$£ = V,,ve, "V $>£ = V ve, 2 2

|0| = m n = *i'V Vn v<»> H .

€<

[.^rifed February, 1921. -The formulae analogous to

for YctYfiy are here greatly generalised and reduced to a form in which they are easy to remember and to apply in particular cases.

In accordance with (3) of § 3 let

| [UV + (-) Vu] = (Ya+b + Ya+b-4, + Va+ft-s + .) UV = AllV 1
ab

.

J[w- (-) ab vu] = (Ya+b -2 + Ya +b-6+ ..>)uv = A'uV

The parts of Aw, and Aw itself, may be said to be concordant with Ya +t>, those of A'uv concordant with Va+ &_ 2. Similar

uvw, as follows,

^(uvw + (-) bc+ca+ab wvu) = Auvw

^(uvw-(-) bG+ca+ab wvu) = A'uvw

be called concordant withVa+6+c and Ya +b+c-2 respectively. [That Auvw as defined by (2) is the same as (V a+6+c + V a+ +c _ + ...) uvw may be proved & 4

from the properties of K.

= ISjwKvKu = -)£ Sa tt+1 >WTO.

X.(1tVW) (

<

Hence 2 A (uvw) = {l + (~)^a+b+cK + b+c +VK}uvw]. a

It is important for present purposes to note the various expression as

(A or A') . u(A or A')vw,

obtained by mere commutations, and especially what are the commutation factors in the various cases. There are two standard cases from which all others may be readily derived. They are

Aim = - ab Auv, ( )

A(Avw . u) = (~-) aib+c) A(uA

and we express them by saying that (--) ab is the commutation factor of Aw, and that (-)a 6+c is the commutation factor for u and Avw in A(uAvw). The

(>

addition of an accent to any A clearly multiplies the commutation factor by - 1. Thus

A'w.= - (-) ah A uv ,

)

A{A'vw u) = - (-) a b+c) A(uA"vw\

.

(

A' (A'vw . n) = -)«<*+«> A' (uA'vw), etc.

(

Thus, If z, y, are independently equal either to
A (r)
A or A' according as x is or 1, and similarly for A \ we have by
for

commutations four forms of A& uA&hw, namely, .

A^(uA^vw) = - y+teAWtyAWwv) (

It is clear that for all present purposes u,
meaning. Hitherto they have meant multenions of the forms Y*p, Y b q, Vcr

but they may here be extended to mean multenions concordant with

that is

v = (Y + Yft+4 + V - +...)?» ^ = (Y + Vc+ +
b & 4

Let now V^ stand for any Yx q, a single selected part, which is con
with Aq ; and Y'q for any single part which is concordant with A'q.
the following 8 statements are identities, and each of the 8 has 64 variants
obtained by such commutations as we have just been considering.
given u, v, w, there are in all 512 variants, and in each of these V, or V, has
in general a multitude of values. In each of the 8 identi
together the individual parts indicated by the V or Y'
or V may in every case be replaced by A or A' respectively.
reference forms those written seem preferable. The ide
(not separately, but taken all together), though not in form, symmetrical in
it,, v, w. This will appear in the course of the proof given later.

Y {Auv . w) •=. Y (uAv<w-\-(-~) ab vA'u,w)

- V {uA'vw + - ab vAuvj), ( )

Y (A.'uv . = Y (uAvio - - ab vAvjw)
•id) ( )

= V('d'w-(- y>vA'uw),

w) = V (uAvio 4-
r
Y (Auv
/
.) ah vAuio) (

= Y (uA'vw -f f ab
vA uw), (•)

V {A'uv . w) = V (t/Am/j-(» ab
vA'uw) •)

= V {uA vw--{ -) vAmu)
ab /

The following remarks will probably render these easy to remember and apply: (1) Counting the accent of a V only once when equation, the number of accents in each of the eight equations of three terms is odd, that is, it is 1 or 3. (2) In these standard forms the sequence u, v is adopted as far as practicable, that is to say, wholly in the left and the first term on the right ; whereas in the second term on th v is forced away from the middle position, the sequence u, w is still maintained. (3) When these rules of sequence are adopted the signs to be given to each term are specially easy of determination ; when to sides of the equation have the same sequence, or by commutation are brought

to the same sequence, they have or are brought to have the when on the same side to have opposite signs. Thus, as wri term on the left and the first term on the right, having the same sequence, have the same (plus) sign ; and again, the term on the left and the second term on the right are brought to the same sequence by commuting u and v on the left., This last determines the sign, +(-) ab in the second term on the right.

We will now throw the eight separate cases of (5) into one form, and V and "V by A and A'. Let g, x y, z each independently henceforth replace t

be equal to or 1. Then all the cases of (5) are included in

A^> (A&uv . w) = Ate) (uAWvto + (- y +ab vA ^ uw)

provided g + x + y-\-z = 1 or 3

Put x', y\ z for the commutation factors of A^viv, A^wu, A^hcv respectively, that is

x' = (- x+bc) , y = -)y +m (, Y= -

It is easy from (3) to prove that -y z\ -z'x\ -x!y are the commutation f f

factors for u and A^ x) vw in Ate) uA^vw and for the like single symbol and . ;

coupled pair in Ate) . vA^wu and Ate) wA^uv respectively. .

To see that (6) is symmetrical in u, v, w, make the commutations necessary to obtain the cyclic order u 9 v, w throughout ; first with the the first place throughout, and secondly in the last place throughout. put all the terms on one side of the equation and write the three terms in the order for which in the first term u occupies the middle position, second v does so, and in the third w. Notice that the coeffic thus come in both arrangements to have their alphabetic order.

AW(x'ioAMi//v + y'tiAWvw + z'vAWiou) = 0,

A^ix'A^wio . v + y'A^'iw w + z'A&>vw
.

To show that these are equivalent to (6), we have only to verify for any two terms in each equation that when brought to the same sequence they have opposite signs, and this is quite easy to do. [Commute w with first term of (8) ; it comes to agreement in sequence and disagreement in sign with the second term.]

We will now prove (8). In the first term in (8), -x'y
f
tion factor and z is the other. Hence four times this term is
4t%' A^wAWuv = 2x (wA< uv- x'y' A^uv z)
. w)
= 2(x'wA(hw-y'A(z
w)

= x' ivuv-\-z'x' wvu~-y'iivw-y'z'
.

We see by cyclic change from this that four times the second and third terms are
y
f
. uvw + x'y' . uwv-z' . vvm-z x' wvu

and z
r
. vivit + y'z' wiw-x*
. . vmv-x'y'

The sum of these three expressions is identica
inspection.

We can reproduce the equations (5) in other notations. Thus, let
mi = Guv, wvu = Cumv,
Ki+C)() = o(), i(i-c)() = C(

Then, similar to (6). the following is true

(>> (C<*>w . w) = C^ (^T3<*>w? + (-)tovCtomo)}^

provided y +x + y-\-z = 1 or 3

The application of the rule of signs is rather improved, but in the reverse is decidedly the case in the application of the proviso. within the brackets each C may be replaced by K, or, instead, by Q, but outside the brackets 0^ must be retained ; also with K o
z

z of (-) must be changed to

z + $[a(a±l) + 'b(b±l)]

the upper sign being taken with K and the lower with Q.
(10) is more closely analogous than (6) to the quaternion VaV/37, but we must remember that in quaternions

a/3 + /3a ~ 2Sa/8, */?7 + 7/3a = 2Ya

ot/3-/3ct = 2Va/3, a/37- 7/3a = 2Sa/37,

so that inside the brackets 0, C are analogous to are analogous to V, S.

Of these different forms (5) is without doubt the best for general use, but the others are occasionally convenient.

Putting A- A' = L and Im = u

we have that 0, K, Q, are all strictly retroseriptive that is
Cuvw = CwCvGu, Kpqr = TJLrK.q'Kp, etc.,
but L is not, though it has the analogous property
Jmv = (-) ab LvLu, Tlmvw = - ( )
bc + ca

These last properties are easily extended to any number of symbols, such as
u, v, w. To pass to higher numbers than three with (5), or say (6), is much
more troublesome. Here is an example of passing to four symbols.
be multenions concordant with Ya Y ', b >. In A^> {A9 htv

outside the bracketsA® is concordant with Va +&+«'+&'-2t,3 first treat ASh\'u' as
a single symbol, and secondly treat A^uv in that manner. We obtain
A<0(A<A>wA<A '>*;V
= A<0(^A<*> vAMv'u' + (-) h+ab v AM uAMv'u'

= A<*>(A<*'> [A^uv ^]tt + (-)*' + «'*'A<y) [A<^m;

provided both + + y)
g + (h + h') (z

and # + (h + A') + (x -f ?/) are odd

In the equation of four terms provided by (11) there would be thirty- two
cases corresponding to the eight of (5), because double value sare assignable
independently to y, h, h' x, x\ In each of the equations of three terms provided by (11) there would be sixteen cases.]

§ 8. Integration. - Let xiii + x&2+ ...+xnin = p stand for an independent
variable position vector of a point in Euclidian space of n dimensions.
point trace a curve of which an element is dp (= dpi).
and trace a two-dimensional tract. In the process let an element of the path
of the element dp (= dpi) be dp The element of the two-dimensional
2.

region traced hereby is given by Y dp dp Let this be continued till a (b + 1)-
2 x 2.

dimensionai tract has been traced.

Let us consider a space integral over the complete boundary (a 6-tract) of the (6+l)-tract and express it as a space integral over the (6+l)-tract. do this, starting from any point of the (&-fl)-tract, let an elementary closed &-traet expand till it has traversed the whole (5 + l)-tract by reaching its boundary. The process could be described in much more detail. the (6-j- l)-tract to be eventually filled with parallelepipedal elements. for each such element the connection between integral over boundary (of element) is expressed as a multiple of Yb+1 dpidp 2 element of (6 + l)-tract. These are summed and in this sum contributions from the common boundary of two elements cancel. The direction indicated by dpb+i is to be into the region bounded by the element

dp b = Ybdpidp 2 ...dp b .

With these conventions the following is true

jwj^fr = f-^&v^+iv*

n

where V = S- i D %

c 1

c ,

$dp& is the element of boundary-integral due to the element of boundary dpb) and dp b +i is an element of the (& + 1)- tract. The form of

of position. The suffix g means that y acts on this function

We will now put the integral (1) in a second form. Let v = XJdp n (Udpn Tdp n = d/5», (Tdp n)2 =

± (dp n) 2 Tdp n a positive real scalar). Also let

}

dpb = vds n - d? is thus a positive scalar, which measures an element

b.

of hyper bulk. d$i is a vector which measures an element of hyper area.

Write db for $?o and da for d^ ; then

db = Yodadpn I

dp n = Y n dpn+ idpn

r- ,

follows from \

Since ^5 is positive and d/? n points inwards, da points outwards in harı

with usual convention.

Equation (1) in the ^9 notation becomes

"" fl)

lM»-« = f

<"" a+1)

f

(

J^Varf^, V,.

The extreme cases of (1) and (4) are when b = 1 and a = 1 that is

J>ip = JJ^Vi^aV^

JC-DJ^« = J«J^v^

In (1) put (f}dp b = a scalar = Yovdp b and make simil
other three. Thus (1) and (4) become

J<»JVo^ ft - (6+1) |Vo^6 + iV6 + iV^
|
0+1
J<»-«>JVoW*d^-=J<»- >JVo*a-lVa-lV(^0»

wZ? has been written in place of a mere u for subsequent convenience,
supposed to be a given scalar function of position of the nature of density.
Thus, if by a general strain p becomes p\ dp becomes dp\ and db becomes db\
then k becomes M where

kdb = Vdb\

k is used in the transformation of (6). (5) and (6) become

lYotrdp = JJVo^ 3V ver, a

J<»-i)JV Tfa*a = JWJ VoV (*t) ^-

We are about to apply these theorems to differential invariants.

surprise some readers that there is so much concerning invariants which is

independent of any conception of differential quadratic f
acquainted with all the theorems here given to the end of § 10 years before I
had heard of quadratic differential forms. My first acquaintance w
forms was after the MS. treatise from which I extract them had been
completed. That first (and up to this date my only) acquaintance with such
forms was through Prof. Wright's delightful Cambridge tract on the subject.
How then did they originate ? From a study of Maxwell's suggestions concern-
ing intensities and fluxes. Till the last three months I never looked at them

from any other point of view. I gave elaborate applications of their three

dimensional form, twenty-seven years ago, to electrical problems.*

§ 9. Oovariants and Contravariants.-Lst p be a vector function of the
independent vector p. We may think of this as representing a displacement
of every point of a medium filling our ^-dimensional Euclidean space from
the position p to the position p\ or we may think of it in its pure mathe
matical aspect as a mere mathematical transformation of the coordinates
xc = -Yoicp into the coordinates xcx = - Yoicp\

Corresponding to an arbitrary increment dp of p, there is an increment
dp" of p* given by

dp = - V rfpv •

P = X d P-

We will use % to denote the linity extended from the vector linity % which
is given by (1). Invariance has to do with relations which remain unchanged

s

in form, in some defined sense or senses, when

p is taken as the independent
variable instead of p. Suppose h is a scalar function of p. Then it is also a
function of p\ Then

dh = -Yodpyh = -Yodptfh,

where v' is $\sum_{c=1}^{n}$ ic D %\ This is a relation of the kind just mentioned and we

want to systematise the treatment of such relations. Any
my paper of 1892 will see how it came about that it should seem a very
natural course to use the theorems of integration of §
purpose.

We scarcely need to analyse the details of the strain after what has
already been said. The ordinary notions of curl, Y 2 V<t, of an intensity, that
is a covariant vector, or, are given by (10) of § 8. The ordi
convergence of a vector flux, far, are given by (11) of § 8. [hr is a vector
flux t is a contravariant vector.]

; The ordinary notions of the half curl,
| Y Y<r, representing the rate of rotation of a fluid whose velocity is
2
just the same in n dimensions as in three [see (10) of § 4 a
said above about infinitesimal rotations]. -J

But more :

-

* ’ Phil. Trans., 5
A, 1892, pp. 685-780.

anybody who has seen fluxes and intensities used as a mathematical method that there is no change in the same uses of the same expres convergence, etc.) when we pass to an ^-manifold of the most general kind. V2v^ and V v (&t) will have just the same significance in the manifold as in the original Euclidean space ; but this is anticipating.

Synopsis of Notation.

Covariant. Contravari

(1) Vector
(2) Hyperyector .... *●●»*'
(3) Multenion
(4) Multenion Unity *** ●●* i * t

We have given an instance of an invariantive statement in (2). consider the integrals (10) and (7) of § 8, which we here reproduce,

\Y$adp = ||V ^p2V2 V<r "1

p>JVo^ = p +1 >JVo^ + iV^V^J

When p" is the independent variable, precisely these same integrals are to be expressed in the same form, thus :

PfVoiM# = J<6+1 >JVod/>6+i Y b+1 M

Now, of course, we are compelled to identify our present dp and dp with the dp and dp" of (1), so that dp" - x^p- Again, dp b meant Y b dpi dp 2 ...dp b where each dp c again must be identified with the dp of (1), and therefore the corresponding dp" with the dp" of (1). Hence

c1
p = xdp> clpt> - x dpt»

by the properties of extended vector Unities.

When we say that the integrals (4) are the same as the integrals (3), we

assert that each of the four elements of integration in (3)
corresponding element in (4). In other words, we assert the exis
of four invariants,

Yoadp - Y adp\ YodfaYjytr = Vo<WV 2 vV

Y vdp = Y vdp \ Vodpb+iYb+iV» = YQ dp b+ i Yb+irfv

b Q b

From (5) and the fundamental property of the conjugate of % tnese at once
give

\ / -I

>

V lv

X >

(5) and (7) suggest multenions of types r, s, such that

r
s
= x r, s" = x ~\f

s s
r and 5 are multenion functions of p, and r and s are associated

(say of p') which produce the same integrals when p
independent variable in place of p [see the synopsis above].
gives the rule of the association of the functions,

r and s are said to be eontravariant and covariant respectively.

It is well to place here the obvious consequence of (8)

Y rs = V rV,

that is, YqP2 is always an invariant if one of the two, ^?, q, is c
the other eontravariant.

By exactly similar reasoning (a little complicated by the presence of k), we may treat of the equations (11) and (8) of § 8. For the sake of saving space, I merely state the results, k being given by
(including as particular cases da and db) being as in (11) and (8) of § 8, the following are covariant :-

kv, kdb, kec, led? a

and the following are eontravariant
kdb, t, to, Aj""
1
VoV(*t)» ^^ViVlH

kdb is both covariant and eontravariant. This merely means that it is
= k dV). 1"1
invariant (kdb
x
Eemembering that yh h x ^ when h is a =
scalar, (8) shows at once that, for a scalar covariance
both mean precisely the same as invariance.

The persistent manner in which k thrusts itself into these results tempts one to wonder whether covariant and eontravariant terminology
to intensity and flux terminology. After some experience of using b
am inclined to believe that the accepted covariant and eontravariant standard scheme is superior to the other.

It is not difficult to show that kv satisfies the test of
that is

If in (2) and (3) of § 6 we read % in place of $ we get theorems closel
}

connected with multiplication and composition in the absolute differential calculus. Let va> v h be covariant multenions of homogeneities, a, b. Then

Va + b = Va + 6W&,

is covariant (multiplication). If u a> u hi are similar contravariants then

Ua+b = Va+bMaUb,

is contra variant (multiplication). If u is contravariant (homogeneity a
v is covariant (homogeneity V), then

Ua -b = Va-bUV,

is contravariant (composition) when a > 6. Also

Vb - a = Vb-aUV,

is covariant (composition) when & < 6. Lastly, when a ~ tl

invariant, which is a special case of (9). A special case arises wl
the larger of the two, a, h equal to n.

£, any multenion Unity, is said to be covariant v
covariant multenion from a contravariant, thus

s = fr, s
1

= fV.

Conversely since then
r - ^-^ r = |f~"V,

f""
1
is said to be contravariant. From (8) we at once get

From this, when £ is covariant, its conjugate f is also covariant, and,
therefore, both the self -conjugate part and the skew part of £ are covari
Similarly for ^~ 1 and contravariance. Also, when f is self-con
self-conjugate. If f is a covariant vector Unity, the extended linity is also
covariant. Also, in this case, the rotation-^ V2 (see (10) of §4) JVs^f? is
covariant. To prove this last, we have to show that

Now %Y2^^Y2X ^XT'C [(2) §6]

= V ft'fxr9 [(9) §5]

= V2??? [(15) above],

as required.

Let 01 a-2 ..., <r n be % independent covariant vectors. [They might be taken
as 57&1, v^2 ••• V^« where the Jacobian of the quantities h does not vanish,
that is, where Yn (yh) ^ 0.] From the relations
{n)

1 - VoO"icri = Vocr =   2 cr2   •-

=Y   -Yocr^3-.. ^   2
>,

it at once follows that a 1} cf 2 > ... o~n are n independent contravariant vec
and <r c> &c might just as properly have been denoted by
covariant or contravariant vector can be expressed at once in terms of
these and invariants. Also, from the 2 n vectoriums formed from each set,

any covariant or contravariant multenion may be
Thus the expressions for or, r, v, u are [see (2) of § 5]

n n

c˜l c-1

v = SV,o-(6
> . VotfVftO-W, tt = 2V,,cr<°) Y uYa
.

§ 10. Differentiation. - Our integration theorems have suggested certain systematisations, now to be given in the use of V- We

If w is covariant, one part on the right, viz., the first, is covariant and the other not. A similar statement applies to contra variance, We cannot, as seems desirable, resolve the symbolic vector y into two parts, to which each of these two discordant parts on the right belong.
the symbolic Unity V( )• The first part of the linity, c
V +iS7 w produces a part of sjw of homogeneity one unit higher than that
c

of w, and may be thought of as the ascending derivative linity, and will be denoted by H. The other part, L, produces a lower homogeneity, and may be thought of as the descending derivative. Thus, formally
Unities H and L thus
v() == H + L

n-l n
where TI == S ve+1 vve (),
=
. L= 2 Vc-i. vV ()|

c

Otherwise yw = Hw + Tjw
where Hw = Vc+ iyw,

There are two, wholly separate, aspects of H and L as ope
both a linear operator and a differential operator.
these two operations is the one symbol, w, but in (4), below (where H' is the conjugate of H), in

we have a case where H', as linear operator, af
succeeding symbol, dp b +x, while as differential operator
is indicated by suffixes in the usual manner. This examp
of the facilities of operation provided by H.

In considering the fundamental properties of H and L,

by a, ctw, Yc+iuw, Yc-io

where a is an ordinary vector instead of a symbolic vector.

:

be the conjugates of H, L as usual. From the func
Yap<f>q = Y(,q<fi'p, we at once have

H'w = Y -iwy, c Uw = Vc+iwy.

Thus the integration theorems (1) and (4) of § 8 read

|W|^ = |(^D|^H/^ 6+1)6

and (7) and (8) of § 8 read

JWJVo*k#» = J<

8+1

>JV dps+1 H»,

]<«-«> Jwfafc a = J(»-« +1)JVod*«-iL(fat).

Further we have

H = =L 2 2

,

and by taking conjugates we have

H' 2 = = L' 2 .

2

From (8) consider the various powers V ( )> V 3 ( ) °^ V ( )•

V 2 () = HL+LH

V ( ) = HLH + LHL, eto.

3

and in (9) we may take conjugates [noting that, since the conjugate of v ( )
is ( ) \7, the conjugate of y 2 ( ) is ( ) y 2 , etc,].

Notice that H' lowers the homogeneity, while 1/ heightens it, so that we
have the following Table :•

Heightens homogeneity.

homogen

Prefixes v H

Postfixes V 1/

Either of the two,H and L, can he expressed in terms of the other thus

Lq = B.(qi) {,~\ TLq = L(qi) v~\

. (10) .

Certain identities satisfied by H and L are given by the theorems of integration. As an example, I dp" = for a closed expressed in terms of p thus

= \X P = cl

l\ Xg

VidfaVg.

Since the whole tract may be taken as the element dp 2 we get

X,Vi4»V, = 0.

If we operate by Vo7( ) we get

or since dfa is arbitrary,

V2V %7 = 0.

Both (11) and (12) are simple enough to be established directly from the form of x> namely, xi = --Vo7V p\ t> u ^ ^ ne method of deriving them, just

•

given, suggests how to prove other results not nearly so easy to establish directly. Instead of jdp

s

= 0we may take either of

and modify them in the same way. Thus from the first, = (6+1

From this and by taking conjugates we get

Hfa's = 0,

X,H/ 2 = 0,

where q is an arbitrary multenion. If we put q = w (13) becomes

XgVc-lWVg = 0, Vc+iVgXgW = °-

Similarly we have

m is here &/&\ £e., it is the Jacobian of p\ or by (11) of § 7

m = n'Y Yn v Yn n (n)

e< >.

Thus = J^-^^fCmx -^ V*«-i. 1

Hence, and by taking conjugates

Putting q = w we get

(™x'

_1

W 2 = °. Lp(™*T V= o-

(^x'^Vc+i^v* = °> ^ViV^X")^ = 0. 1

Let us now find expressions explicitly giving v\ H\ L in terms involving v

V, H, L. By (2), § (9), v is a symbolic covariant vector, so that

V^ = X'-'V,

where on the right the differentiations are not to affect x~ l

-

For H we have that Hs is covariant when s is. Hence

HV = % '-iHS = % '~iH(% V).

Now, by the second of (13) in this expression we may show the differentiations of H explicitly as affecting or as not affecting the immediately following X\ because the terms implied by those differentiations are identically zero. Hence we may write

H' = x '-i . H . x ',

bearing this permissibility of interpretation in mind. The relations